

Early-Warning Signals of Regime Shifts in Euro-Area Sovereign Spreads: A Complexity-Theory Approach

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Abstract

This thesis investigates whether tools from complexity theory can provide useful early-warning signals (EWS) of regime shifts in euro-area sovereign bond markets. Focusing on the Spain-Germany 5-year benchmark yield spread, it constructs a set of complexity-based indicators within a rolling-window framework, including measures of critical slowing down, autocorrelation, variance, skewness, and flickering, and evaluates their behaviour ahead of five discrete crisis episodes drawn from the euro-area sovereign debt crisis of 2009 to 2013.

The results are broadly supportive but nuanced. For three episodes driven primarily by endogenous deterioration in Spanish spread dynamics: the first ECB Securities Markets Programme intervention, its expansion to Spain/Italy, and the Spanish banking-sector assistance request, the EWS signals are present and meaningful ahead of each anchor date. Concentration was particularly pronounced in the days immediately preceding the SMP launch and the banking-sector assistance request, and distributed across longer horizons ahead of the Spain contagion episode, consistent with its more gradual build-up. For the Greek fiscal shock, the near-total absence of EWS activity is interpreted as an informative baseline rather than a methodological failure: Spanish spreads were not yet in a pre-transitional state, and the indicators reflected this accurately. For the LTRO episode, the similarly limited EWS activity is consistent with the intervention being a discretionary policy decision rather than the outcome of endogenous spread fragility; the indicators correctly did not signal a critical transition that was not occurring.

Statistical analysis of spread dynamics following EWS dates reveals that the indicators function as markers of elevated systemic fragility rather than as directional predictors of further spread widening. These findings are cross-validated against a Markov-switching autoregressive model, which provides partial but imperfect statistical support for the hand-selected chronology. The thesis concludes by discussing the limitations of a single-maturity approach, the noise properties of the 5-year tenor, and the scope for extending the framework to multiple maturities and other peripheral sovereigns as directions for future research.

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1 Introduction

1.1 Complexity Theory and Financial Systems

Complexity economics represents a shift away from traditional equilibrium-based frameworks toward viewing the economy as a dynamic, evolving system driven from the bottom up (Beinhocker et al. 2026). Rather than assuming that markets tend toward a stable equilibrium, this approach emphasises the role of heterogeneous, interacting agents whose behaviours continuously adapt in response to new information and changing conditions. In this setting, macroeconomic phenomena such as financial bubbles or clustered volatility are not treated as the result of exogenous shocks, but as emergent outcomes arising from the repeated interactions of these agents over time. Because agents operate under fundamental uncertainty, they cannot optimise perfectly and must instead revise their beliefs and strategies iteratively, generating a constantly evolving system (Arthur 2013).

Proponents of complexity theory argue that this bottom-up perspective is essential for understanding real-world crises, especially in systems characterised by strong interconnectedness and feedback effects (Battiston et al. 2016), where shocks do not remain isolated but instead propagate through networks and become amplified via contagion and feedback mechanisms, turning local disturbances into system-wide instability. From this perspective, many financial crises are not driven solely by exogenous shocks, but by endogenous processes that build up gradually within the system (Battiston et al. 2016), implying that a gradual loss of resilience can leave the system increasingly fragile, such that even a small disturbance may trigger a disproportionately large breakdown. For this reason, identifying early warning patterns during periods of turbulence may help researchers and policymakers better understand how crises develop and how major regime changes emerge.

As the degree of connectivity within the system increases, the economy may approach a critical threshold beyond which its behaviour changes qualitatively. At this point, the system can undergo a phase transition, shifting from a relatively stable regime to one characterised by fragility and large fluctuations. Central to this idea is the concept of a “tipping point” or bifurcation. Borrowed from ecology, a tipping point refers to a situation in which a small change in an underlying parameter, such as interest rates or market sentiment, produces a disproportionately large and potentially irreversible shift in the state of the system (Hastings et al. 2026). These abrupt transitions occur because the interconnected nature of the networks themselves serve to amplify shocks: what begins as a minor disturbance can explode through the system in a domino-like fashion (Battiston et al. 2016).

From a mathematical perspective, such transitions can be understood in terms of a loss of stability in the system’s underlying structure. Specifically, when a critical manifold loses its normal hyperbolicity, stabilising forces that would ordinarily restore the system to equilibrium weaken or disappear (Kuehn 2011). Intuitively, this can be thought of as the system losing its “restoring pull” or “gravity” such that deviations, like widening financial spreads, no longer self-correct, leaving the system increasingly fragile and susceptible to collapse following even small shocks. This framework highlights how gradual changes in structural conditions can push the economy toward a tipping point without any single large shock being necessary.

Importantly, before such a transition occurs, complex systems like financial markets often exhibit identifiable early-warning signals. As a system approaches a tipping point, it tends to recover more slowly from disturbances, a phenomenon known as critical slowing down (Scheffer et al. 2009). This is typically reflected in changes across a range of statistical indicators, the precise definitions and construction of which are outlined in detail in Section 2.2. Together, these measures provide observable signals that the system’s resilience is deteriorating.

Building on this insight, in this paper we aim to investigate the reliability of these early-warning signals in assessing the likelihood of an impending regime shift. The work of van den End (2019) serves as the foundational framework for this proposal. The paper uses complexity indicators of autocorrelation, variance, skewness, and critical slowing down to identify critical transitions within the interest rate market following the financial crisis. van den End (2019) argues that massive injections of excess liquidity fundamentally altered market functionality, creating positive feedback loops where market participants, increasingly dependent on central bank support, reduced their own risk protection. This process erodes systemic resilience, making the market more prone to abrupt regime shifts.

While van den End (2019) focuses on interest rate dynamics, Metadger and Raghavendra (2013) demonstrate the efficacy of these indicators specifically within sovereign debt markets. They provide empirical evidence that the Eurozone sovereign debt crises in Greece, Ireland, and Portugal were preceded by critical slowing down (Metadger and Raghavendra 2013). Their assertion that critical slowing down is a universal property of systems approaching a tipping point validates the application of these metrics to Spanish spreads and supports the hypothesis that they can serve as useful early-warning signals.

However, the optimism of van den End (2019) and Metadger and Raghavendra (2013) must be balanced against the limitations of these indicators. Guttal et al. (2016) caution that financial meltdowns may not always follow the mathematical path of a classical critical transition. In their analysis of various stock market indices, they found no evidence of critical slowing down preceding major crashes, suggesting that financial markets may move too rapidly or follow different dynamics that do not produce these specific signals (Guttal et al. 2016). Furthermore, Boettiger and Hastings (2012) suggest that rising variance and autocorrelation can occasionally emerge as consequences of large fluctuations in stable systems, rather than underlying structural changes. Therefore, caution must be taken before drawing any solid conclusions from the early-warning signals as changes in the variables can be highly unreliable when transitions are driven by chance, potentially leading to false positives.

Nonetheless, despite these challenges, Guttal et al. (2016) did also find that rising variance does indeed precede many financial meltdowns, suggesting it may be a more robust predictor of systemic risk than autocorrelation or critical slowing down. This justifies our decision to test a range of multiple indicators to capture diverse market stresses, as not every financial transition will behave like a textbook bifurcation. However, testing early-warning signals requires selecting a market variable that captures sovereign stress with sufficient sensitivity and theoretical coherence. This motivates the choice developed in the following section.

1.2 Sovereign Spread Dynamics in the Euro Area

Having established that complexity-based indicators can in principle detect the build-up of fragility in a system, the question becomes which market variable best reflects that fragility in the euro-area sovereign context. A natural starting point for measuring sovereign stress in the euro-area is the yield spread between a country's sovereign bonds and a safe benchmark. Given the central role of German government bonds as the reference security in European fixed income markets (Favero and Missale 2012), this thesis focuses on the Spain-Germany spread at the 5-year maturity. The rationale behind the selection of the 5 year maturity specifically will be discussed in Section 2.1 of this thesis.

Spain represents a particularly instructive case for this analysis. As the fourth-largest economy in the euro area, it was large enough that its sovereign stress carried genuine systemic implications for the monetary union as a whole, yet unlike Greece it did not require a full sovereign bailout programme. It experienced a full crisis arc: a sharp deterioration followed by stabilisation, making it well-suited to testing whether early-warning indicators can detect the build-up and resolution of spread fragility across a complete regime cycle.

That said, yield spreads alone do not capture the full character of sovereign stress. Garcia-de-Andoain and Kremer (2017) show that stress also manifests through higher volatility and weaker liquidity conditions, proposing instead a composite indicator that combines all three dimensions. This broader view matters because liquidity and credit risk do not always move in tandem: a widening spread may primarily reflect impaired market functionality rather than a genuine deterioration in creditworthiness, and the distinction carries direct implications for policy. If stress is driven by illiquidity, interventions should target market functioning; if it stems from a credit shock, a backstop instrument is more appropriate (Gómez-Puig, Pieterse-Bloem, and Sosvilla-Rivero 2023). Collapsing these dynamics into a single spread figure risks misreading the nature of the tension.

This measurement challenge becomes even more acute when sovereign bond markets are considered in an international context, since stress rarely remains contained within a single country. Sovereign bond markets are interconnected across borders and maturities, and shocks transmit through these linkages in ways that can intensify domestic conditions. Rout and Mallick (2022) document that spillovers in sovereign bond yields strengthened markedly during the Covid-19 period, particularly across medium- and long-maturity bonds, with short-maturity spillovers remaining lower; consistent with the view that long-term rates are more exposed to global investor dynamics. Crucially, Garcia-de-Andoain and Kremer (2017) show that these spillover patterns are time-varying, and that using a plain yield spread rather than their broader aggregate measure can generate a more limited picture of sovereign stress and the transmission of tensions. The structure of contagion, therefore, depends on how stress is measured in the first place.

These spillover dynamics take on particular significance when sovereign bond markets transition between regimes. In the euro area, such transitions have historically been abrupt and asymmetric: during the early years of the monetary union through to around 2009, spreads were largely explained by macroeconomic factors, with market liquidity and global sentiment playing secondary roles (Eijffinger and Pieterse-Bloem 2023). The European debt crisis of 2009-2012 broke this pattern sharply. Shocks to

fiscal deficits drove pronounced divergences in Eurozone spreads, and sovereign risk began to be priced differently across the monetary union (Eijffinger and Pieterse-Bloem 2023), marking a clear divide between core and peripheral countries. This episode also showed that the academic literature on macroeconomic determinants of spreads, which had mostly focused on tranquil conditions, provided limited guidance on behaviour during actual crises, when the pricing of sovereign risk can deviate substantially from its usual relationship with fundamentals (Blommestein, Eijffinger, and Qian 2016).

Beyond fiscal fundamentals, sentiment and communication factors have also been shown to influence spread dynamics, and these tend to interact with the broader regime in non-trivial ways. Georgoutsos and Migiakis (2013) identify economic and market sentiment as among the most significant drivers of sovereign spreads, while Gade et al. (2013) show that political communication exerts a measurable effect: positive signals from policymakers compress spreads, whereas negative or disorderly communication widens them. What these findings collectively suggest is that spread dynamics are not governed by hard macroeconomic data alone, but also by how market participants interpret and react to signals; a process that can become self-reinforcing during periods of elevated uncertainty, as investors rapidly reduce exposure to credit-sensitive securities and perceived default risk spirals away from fundamentals (Costantini and Sousa 2022).

It is precisely in these circumstances that institutional interventions become most consequential. In summer 2022, the ECB's shift towards monetary tightening contributed to a renewed widening of spreads in the most indebted euro-area countries, rekindling concerns about fragmentation within the monetary union (Gómez-Puig, Pieterse-Bloem, and Sosvilla-Rivero 2023). The ECB responded by introducing the Transmission Protection Instrument (TPI) in July 2022, which permitted secondary-market purchases of sovereign bonds when financing conditions deteriorate in ways not justified by country-specific fundamentals. Its introduction reflected a recognition, consistent with the earlier discussion of liquidity versus credit risk, that the appropriate policy response depends on correctly diagnosing the source of stress: the TPI addresses credit-driven fragmentation, but if stress originates in liquidity conditions, the instrument is less well-targeted (Gómez-Puig, Pieterse-Bloem, and Sosvilla-Rivero 2023).

Taken together, this body of evidence suggests that sovereign spreads are shaped by an interplay of fiscal fundamentals, market sentiment, international spillovers, and institutional interventions, and that the relationships between these factors are neither stable nor linear. Regimes can shift suddenly, the dominant source of stress can change within an episode, and simple spread-level indicators may lag these transitions. This is precisely the gap that tools from complexity theory aim to address. Early-warning signals may detect the build-up of this instability before it fully materialises in spread levels. The remainder of this thesis develops and tests this approach, with the aim of showing that complexity-theoretic indicators can offer useful advance notice of regime shifts in Spanish sovereign spreads.

2 Data and Methodology

2.1 Data

Our empirical analysis is based on daily benchmark government bond yield data downloaded from LSEG and subsequently analysed in Python. We use benchmark yields because they provide a consistent and market-recognised representation of sovereign borrowing costs at comparable points of the yield curve. In the LSEG benchmark bond framework, the benchmark series is not an average of all bonds in a maturity bucket; rather, the benchmark bond is selected as the bond considered most representative for a given maturity band, with selection based on market convention and factors such as yield, liquidity, issue size and coupon (LSEG Datastream, [n.d.](#)). This makes benchmark yields a suitable choice for cross-country spread construction, since our objective is to compare Spain and Germany at equivalent maturities.

The dataset contains daily benchmark government bond bid yields for Spain and Germany at the 5-year, 10-year, and 30-year maturities. After importing the data, the `Date` variable is converted into a datetime index for easier handling, the observations are sorted in chronological order, and rows with missing values are removed. The resulting cleaned series contain 6,834 daily observations for each country and cover the period from 1999-12-31 to 2026-03-11.

For each maturity $m \in \{5Y, 10Y, 30Y\}$, the sovereign spread is constructed as

$$\text{Spread}_t^{(m)} = y_{\text{Spain},t}^{(m)} - y_{\text{Germany},t}^{(m)}.$$

The selected sign convention is methodologically important: Germany is treated as the euro-area benchmark issuer, so an increase in $\text{Spread}_t^{(m)}$ is interpreted as a deterioration in Spain's relative sovereign risk conditions. Although the full sample is available, our main empirical analysis focuses on the euro-area sovereign debt crisis, since this episode represents a clear period in the sample in which Spanish sovereign risk was strongly repriced relative to Germany. Accordingly, the spread data are restricted to the date range from 2000-01-01 to 2013-07-31. This window retains a sufficiently long pre-crisis period for the estimation of rolling indicators while keeping the analysis centred on the build-up and escalation of sovereign stress.

Furthermore, we use the 5-year Spain-Germany benchmark spread as the main state variable. This choice is partly practical, but it is also grounded in its role as the market convention benchmark for sovereign credit risk assessment. The 5-year tenor is widely regarded as a relatively liquid and actively traded point on the sovereign yield curve, and therefore tends to provide the cleanest and most reliable signal of sovereign credit conditions rather than idiosyncratic pricing noise (Longstaff et al. [2011](#); Pan and Singleton [2008](#)). This liquidity advantage also helps ensure greater consistency of the data over the full sample period.

It is also motivated by the fact that the 5-year maturity appears relatively sensitive to changes in sovereign risk conditions during crisis periods. Compared with the 10-year and especially the 30-year spread, the 5-year spread may react more clearly to medium-term repricing of Spanish sovereign risk, while being less affected by very long-term duration effects and term premium distortions. Focusing on a single maturity further allows for a cleaner and more rigorous analysis of the core research question, since examining spread levels at a single tenor is a standard approach for isolating common

and country-specific sovereign risk factors, whereas incorporating multiple maturities introduces additional yield-curve effects that would require separate modelling.

During the data-gathering stage, we also inspected the 10-year and 30-year Spain-Germany spreads. These maturities broadly display the same crisis narrative, but the 5-year spread provided the clearest and most convenient series for the main regime-switching and early-warning analysis. The 10-year and 30-year spreads are therefore left as natural extensions for future robustness analysis rather than forming part of the central empirical results. Analysis across multiple maturities remains a natural extension of this work.

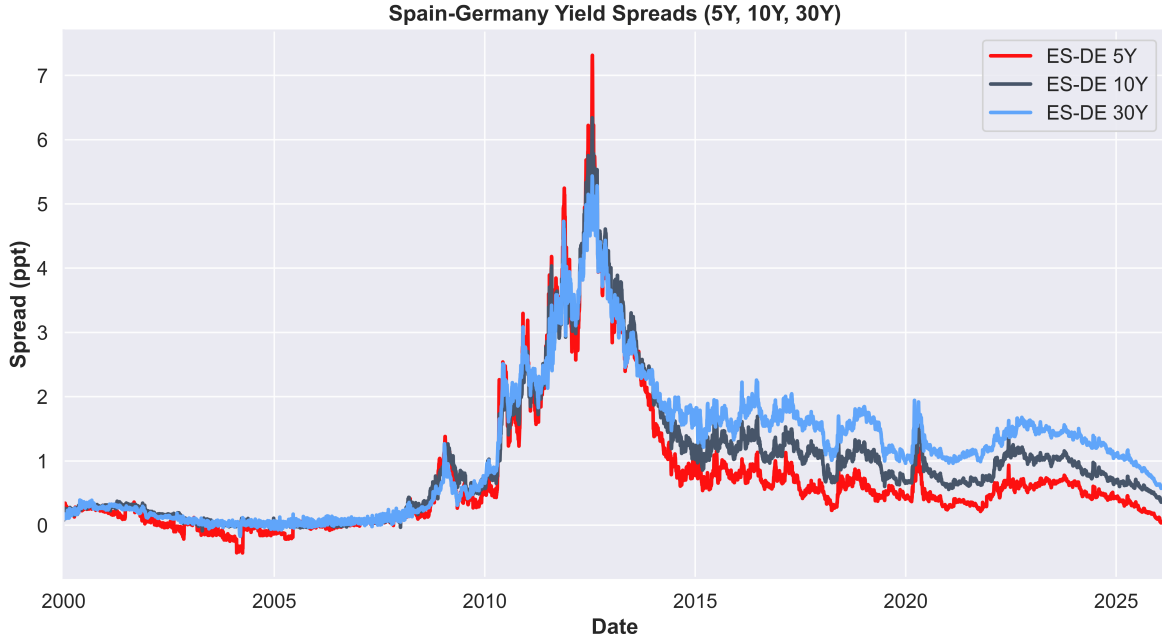


Figure 1: Spain-Germany Sovereign Yield Spreads, 2000–2026.

Moreover, it is important to highlight that some benchmark-yield series display short periods of relatively flat or step-like movements, especially in the earlier part of the sample. This is likely related to the construction of benchmark-yield data. LSEG benchmark series are based on representative bonds within a given maturity band, rather than on a perfectly continuous constant-maturity yield curve. As a result, periods of limited trading activity, stale bid quotes, or changes in the selected benchmark bond can produce movements that appear somewhat mechanical. In particular, benchmark switches may create artificial-looking behaviour: when LSEG changes the representative bond within a maturity bucket, the series can show small jumps, flat sections, or changes in dynamics that reflect data construction rather than pure market movements. Therefore, we keep this in mind for our analysis and treat this aspect of our dataset with caution.

2.2 Construction of Complexity Indicators

2.2.1 Indicators

Following the literature on critical transitions in complex systems, several indicators can be used to assess whether a system is approaching a tipping point. In this thesis, these indicators are applied to the behaviour of the yield spreads under study. As emphasized by Scheffer et al. (2009), early-warning indicators are useful because the loss of resilience in a system often develops gradually before the transition itself becomes visible. In line with van den End (2019), the main indicators considered here are critical slowing down, autocorrelation, variance, skewness, and flickering.

Critical Slowing Down. Critical slowing down refers to a phenomenon in which a system recovers increasingly slowly from small disturbances as it approaches a tipping point. Scheffer et al. (2009) show that this slowing down can begin well before the transition itself, with the rate of recovery from small perturbations gradually declining toward zero. When this rate reaches zero, even the smallest disturbance can send the system into a new regime. Consequently, the speed at which a system recovers from minor disturbances can serve as an indicator of the system’s proximity to a bifurcation point.

Following van den End (2019), critical slowing down is operationalised through AR(1) persistence. The direct decay-rate formulation used by van den End becomes problematic with sovereign spread data where the state variable can take negative values, since the logarithmic transformation it requires loses its interpretation. Instead, for each rolling window the following AR(1) process is estimated:

$$N_t = c + \rho N_{t-1} + \varepsilon_t \quad (1)$$

where ρ denotes the estimated persistence of the spread. The decay rate is then calculated as:

$$\tau = -\ln(\rho) \quad (2)$$

This formulation applies when $0 < \rho < 1$. As ρ approaches one, τ approaches zero, indicating that the spread is recovering from disturbances increasingly slowly: the defining mark of critical slowing down. A low value of τ therefore signals that the system is losing resilience and may be approaching a transition. We therefore characterise critical slowing down by the decay rate approaching zero. While this may seem counterintuitive, larger movements in spreads typically reflect a variable that is still dynamically adjusting like normal and therefore not yet exhibiting slowing down. After a regime shift, it can be common to then observe large increases in this metric, as the system regains dynamism.

Autocorrelation. A rise in autocorrelation is another commonly discussed early-warning signal. As the system slows down, its current state becomes increasingly similar to its recent past, which implies greater persistence in the dynamics of the state variable (Scheffer et al. 2009). Following van den End (2019), this persistence is measured by the coefficient ρ from a first-order autoregressive model:

$$N_t = c + \rho N_{t-1} + \varepsilon_t \quad (3)$$

where ρ is estimated by ordinary least squares. A rise in ρ indicates that the system is becoming more persistent and therefore potentially less resilient.

Variance. Model-based analyses suggest that the variance of the state variable often increases before a critical transition. The intuition is that, when recovery becomes slower, the effects of shocks decay less quickly, so their cumulative impact becomes larger over time (van den End 2019). The sample variance of the state variable N is given by

$$\sigma^2 = \frac{1}{n-1} \sum_{t=1}^n (N_t - \mu)^2 \quad (4)$$

where n is the window length over which the variance is calculated and μ is the sample mean of the state variable. An increase in σ^2 may therefore indicate that the system is moving closer to a critical threshold.

Skewness. Before a bifurcation, fluctuations in the state variable may become more asymmetric. Close to a tipping point, the system may spend relatively more time near the unstable equilibrium, which makes the distribution of the state variable increasingly skewed. Following van den End (2019), skewness is measured by the third standardized moment:

$$\gamma = \frac{m_3}{\sigma^3} = \frac{\frac{1}{n} \sum_{t=1}^n (N_t - \mu)^3}{\left[\frac{1}{n} \sum_{t=1}^n (N_t - \mu)^2 \right]^{3/2}} \quad (5)$$

where m_3 denotes the third central moment, σ is the standard deviation, and μ is the sample mean. Higher skewness suggests increasing asymmetry in the fluctuations of the system.

Flickering. Flickering describes a situation in which the system starts to alternate between the old state and the new state before the critical transition has fully occurred. This behaviour reflects the coexistence of competing attractors in the vicinity of the tipping point. In empirical work, flickering can be captured through the bimodality of the distribution of the state variable. Following van den End (2019), this is measured using Sarle's bimodality coefficient:

$$\beta = \frac{\gamma^2 + 1}{\kappa} \quad (6)$$

where γ denotes skewness and κ denotes kurtosis. The coefficient β ranges between 0 and 1. A higher value indicates stronger bimodality, which may reflect greater instability and more pronounced oscillation between regimes.

Taken together, these indicators provide a set of complementary signals about whether the system is losing resilience and approaching a regime change. In our analysis, they are used to evaluate whether the behaviour of spreads can display early-warning patterns consistent with the approach to a critical transition.

2.2.2 Methodology and Early-Warning Signals

The complexity theory indicators are calculated using a rolling-window approach. For the Spain-Germany 5-year spread series, all five indicators are estimated over 100-day rolling windows. This means that each calculated value of the indicators reflect on the behaviour of the spread over the previous 100 trading days. Also, it is important to highlight that $\tau = -\ln(\rho)$ is only defined for $0 < \rho < 1$, so any window with $\rho \leq 0$ or $\rho \geq 1$ is assigned $\tau = \text{NaN}$ (Not a Number) and the values are not considered in the analysis.

After calculating the rolling indicators, each indicator is compared with its historical distribution. Thus, this is an ex-post study, and the warning thresholds should be interpreted as retrospective cut-offs rather than real-time warning signals. To continue, a warning signal is generated when the indicator enters its relevant 10% tail: the lower 10% tail for the slowdown indicator and the upper 10% tail for autocorrelation, variance, skewness, and flickering. The choice of a 10% threshold is imposed as a modelling assumption and serves as a pragmatic cut-off for identifying unusually extreme indicator values. In this analysis, the 10% threshold is chosen as a fixed modelling choice rather than being estimated from the data.

After calculating the rolling indicators for each window and the corresponding signals, the individual indicator signals are combined into a daily early-warning score. This score is calculated as the number of indicators that are in their warning region on a given date. With five indicators, the score can therefore range from 0 to 5. A higher score indicates that more complexity indicators are simultaneously signalling abnormal behaviour in the 5-year sovereign spread. van den End (2019) also assesses these indicators similarly, using tail exceedances and pre-transition trends to evaluate whether they provide early-warning information before regime shifts in euro-area interest-rate markets.

3 Regime Classification

Our selected crisis dates in Section 4 are based on historical and economic interpretation. In order to support the crisis date selection statistically, we decided to use a Markov-switching model to identify whether changes in the latent regime structure of the Spain-Germany 5-year sovereign spread occur near the selected dates, within our analysed date range.

More specifically, following the regime-dating approach used in Markov-switching models, a two-state Markov-switching autoregressive model is estimated for the Spain-Germany 5-year sovereign spread. The purpose of the model is to classify the spread into calm and stress regimes and thus provide a statistical benchmark for the historically selected crisis dates. This follows the logic of van den End (2019), who uses a two-regime autoregressive switching model to identify regime changes in euro-area interest-rate configurations. The model assumes the spread is generated by one of two unobserved states at any given time, and estimates which state is most likely on each day based on the spread's behaviour. Within this framework, each latent state governs a distinct structural regime characterized by its own mean-reverting dynamics, autoregressive parameters, and conditional variance. The temporal progression between these regimes

is governed by a first-order Markov chain, which quantifies the persistence of each state and the likelihood of structural shifts from one period to the next.

Let s_t denote the standardised Spain–Germany 5-year spread at time t , and let $q_t \in \{1, 2\}$ denote the unobserved regime. The Markov-switching AR(1) model is specified as

$$s_t = c_{q_t} + \phi_{q_t} s_{t-1} + \varepsilon_t, \quad (7)$$

where c_{q_t} is the regime-specific intercept and ϕ_{q_t} is the regime-specific autoregressive coefficient. The disturbance term is assumed to be normally distributed with regime-specific variance,

$$\varepsilon_t \mid q_t = k \sim \mathcal{N}(0, \sigma_k^2), \quad k \in \{1, 2\}. \quad (8)$$

The latent regime variable follows a first-order Markov chain with transition probabilities

$$p_{ij} = \Pr(q_t = j \mid q_{t-1} = i), \quad i, j \in \{1, 2\}. \quad (9)$$

The diagonal probabilities p_{11} and p_{22} therefore measure the probability of remaining in the same regime from one trading day to the next, while the off-diagonal probabilities measure the probability of switching between regimes (see Table 6 in Appendix A.5).

The specification allows both the autoregressive coefficient and the innovation variance to differ across regimes. This is important because the early-warning-signal framework is partly based on changes in persistence before regime shifts. Allowing ϕ to switch across regimes avoids imposing equal persistence across calm and stress states and makes the Markov-switching specification consistent with the critical-slowness interpretation.

After estimation, smoothed regime probabilities are obtained for each observation. Each date is assigned to the regime with the highest smoothed probability. The regimes are then reordered according to the average level of the original, non-standardised spread, so that Regime 1 corresponds to the higher-spread stress regime and Regime 2 corresponds to the lower-spread calm regime.

In the resulting classification, Regime 1 is associated with higher spread levels and greater volatility, while Regime 2 is concentrated in lower-spread periods (see Table 5 in Appendix A.4). The model therefore can be useful as a statistical regime-classification tool. However, consistent with the caveats discussed by van den End (2019), the Markov-switching classification is not interpreted as standalone proof of a critical transition. Instead, it is used as a complementary benchmark alongside the historically selected sovereign-crisis dates and the rolling complexity indicators (see Appendix A for full diagnostics and estimation details).

Looking ahead to the fitted model results, the estimated AR(1) coefficients are near-identical across regimes, which suggests that persistence is a general feature of the Spain–Germany 5-year spread, rather than a characteristic unique to the stress regime (see Table 7 in Appendix A.5). For this reason, the Markov-switching specification should not be interpreted as providing direct cross-validation of critical slowing down through higher stress-regime persistence. This does not undermine the rolling early-warning analysis, since the two approaches capture different dimensions of persistence. The Markov-switching AR coefficients summarise average persistence within each latent

state, while the rolling complexity indicators capture local changes in persistence before specific transitions. Consequently, evidence of critical slowing down is more appropriately evaluated through the time-varying rolling indicators than through the average regime-specific AR coefficients. For brevity, we hereafter refer to this model as the Hidden Markov Model (HMM).

4 Crisis Chronology

In order to verify the effectiveness of our indicators, we have identified a series of key dates that we believe represent the most significant structural disruptions to euro-area government bond markets over our sample period. While we acknowledge that the selection of such dates is an inherently subjective process, we have sought to apply a rigorous and systematic approach by drawing on official communications from institutions including the ECB and the European Commission, as well as corroborating evidence from the academic literature. Priority was given to events widely recognised as structural turning points in the development of the crisis.

4.1 The 2008 Global Financial Crisis

The Financial Crisis of the late 2000s created the structural preconditions for the sovereign debt crisis that followed. The episodes documented here were originally candidates for the early-warning signal (EWS) analysis, but the Spain-Germany spread proved largely unresponsive to the most significant 2007-2008 events, showing little evidence of the regime dynamics the complexity indicators are designed to detect. The absence of spread movement reflects a genuine lack of sovereign market stress in Spain at this time, not a failure of the methodology. These dates are nonetheless retained to provide contextual scaffolding for the subsequent sovereign debt crisis: the transfer of banking-sector losses onto sovereign balance sheets, set in motion by the Eurozone recapitalisation of October 2008, is the direct structural link between the two crises (Acharya, Drechsler, and Schnabl 2014; Reinhart and Rogoff 2011), explaining why Spanish spreads subsequently became so sensitive to domestic bank fragility (Bellia et al. 2019).

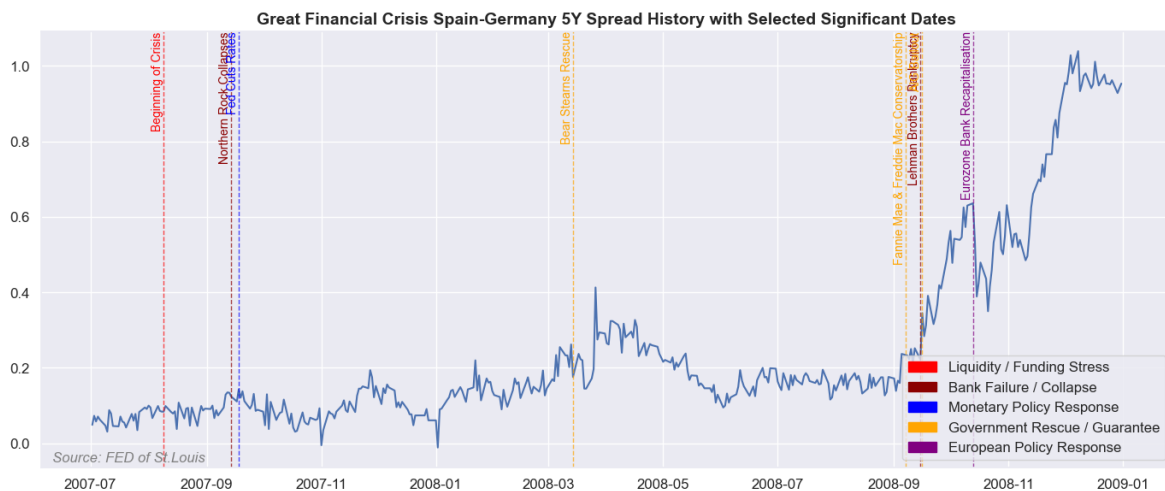


Figure 2: Great Financial Crisis Spain-Germany 5Y Spread History.

9th August 2007: Beginning of Crisis. BNP Paribas suspended fund withdrawals citing "the complete evaporation of liquidity" in certain US market segments (BNP Paribas 2007), prompting the ECB to inject €95 billion into the eurozone banking system (Stark 2009).

14th September 2007: Northern Rock Collapses. The UK's first bank run in 150 years saw £1 billion withdrawn before the government guaranteed deposits (Guillén 2011). Northern Rock's reliance on non-retail funding left it acutely vulnerable to short-term liquidity stress (Shin 2009).

18th September 2007: Fed Cut Rates. The Fed cut rates by 50 basis points from 5.25% (FOMC 2007), beginning an easing cycle that would see rates reach 1.00% within a year (Tepper 2026), materially shifting the global interest rate environment and risk appetite.

14th March 2008: Bear Stearns Rescue. An emergency Fed credit extension prevented Bear Stearns' collapse (Fed Board 2008), establishing the precedent that sovereigns would absorb domestic financial sector losses and thereby linking bank vulnerabilities to sovereign yields.

7th September 2008: Fannie Mae & Freddie Mac. The two institutions, covering nearly half of outstanding US mortgages (Guillén 2011), were placed into conservatorship due to systemic risk concerns (Paulson 2008), further raising questions about Spanish banks' real estate exposure.

15th September 2008: Lehman Brothers Bankruptcy. Lehman's collapse, the largest bankruptcy in history at over \$600 billion in assets (Wiggins, Piontek, and Metrick 2019), sent the FTSE 100 down 4% and the IBEX 35 down 4.5%, marking the transition from localised credit stress to global systemic panic (Guillén 2011).

16th September 2008: AIG Bailout. Up to \$85 billion was lent to prevent AIG's collapse, deemed too interconnected to fail given its global network of exposures (New York FED, n.d.).

13th October 2008: Eurozone Recapitalisation. Coordinated sovereign recapitalisation of eurozone banks (Summit of the Euro Area Countries 2008) restored short-term

confidence (Deutsche Börse, [n.d.](#)) but entangled sovereign and bank balance sheets (Bellia et al. 2019), forming the direct structural mechanism linking the global financial crisis to the eurozone sovereign debt crisis. While the recapitalisation marked a decisive structural shift, its effects on Spanish sovereign spreads took time to materialise. Spreads remained contained through 2009 until Greece’s true fiscal position was revealed, at which point the sovereign debt crisis properly begins.

4.2 The Euro-Area Sovereign Debt Crisis

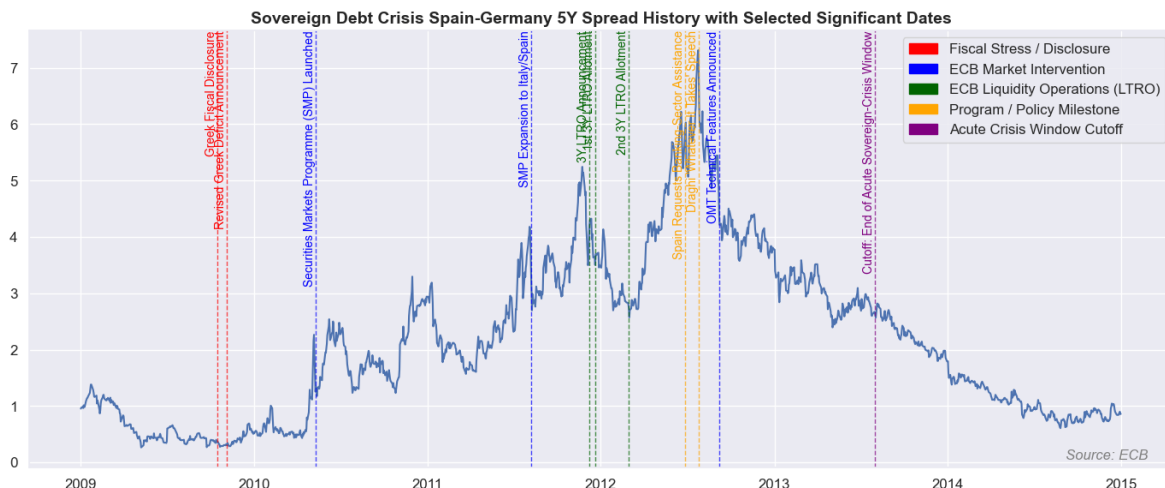


Figure 3: Sovereign Debt Crisis Spain-Germany 5Y Spread History with Selected Significant Dates.

16 October 2009: Greek fiscal disclosure. The Greek fiscal disclosure of 16 October 2009 marked the beginning of the euro-area sovereign debt crisis. In his first parliamentary speech, Prime Minister Papandreou publicly revealed that the severity of Greece’s fiscal position was substantially worse than previously expected. This date is an early inflection point in the crisis (De Santis 2012), following which there was a structural break in the pricing of peripheral sovereign risk, as EMU membership was no longer seen as an implicit guarantee against default (Arghyrou and Kontonikas 2012). For Spanish spreads, the disclosure introduced a new risk dimension: the possibility that fiscal misreporting was not confined to one country and that the eurozone’s institutional framework might prove insufficient to prevent sovereign default.

5 November 2009: Revised Greek deficit announcement. The revision of Greece’s 2009 deficit to 12.5% of GDP, roughly double the previously reported figure, was the first quantitatively explicit fiscal shock of the crisis (Sibert 2010). De Santis (2012) documents a sharp rise in sovereign spreads across the euro area following this announcement, as it directly undermined the credibility of EU fiscal surveillance mechanisms. Lane (2012) treats late 2009 as the moment the crisis entered its active phase, noting that several other euro-area nations soon disclosed deficit-to-GDP ratios far exceeding earlier projections. For Spain, this event reinforced concerns about the deteriorating public finances caused by the unwinding of the property boom.

10 May 2010: Launch of the Securities Markets Programme. The SMP was the ECB's first direct intervention in sovereign bond markets during the crisis and involved the direct purchase of government debt from distressed countries to ensure depth and liquidity within markets (Deutsche Bundesbank, [n.d.](#)). The programme sought to directly address the inherent vulnerability to self-fulfilling liquidity crises that arose from eurozone governments issuing debt in a currency that they cannot control (De Grauwe and Ji [2013](#)). Lane ([2012](#)) interprets the ECB's purchases as a mechanism for preventing a bad equilibrium by restoring liquidity to dysfunctional sovereign-debt markets, and Ghysels et al. ([2017](#)) laud the SMP's success in doing so, reporting that the announcement alone caused ten-year Greek spreads to fall by more than 400 basis points on the day.

7 August 2011: SMP expansion to Italy and Spain. Prior to this intervention, sequential bailout requests from Ireland (November 2010) and Portugal (April 2011) had confirmed that the crisis was not confined to Greece. Spread dynamics across this period increasingly reflected contagion and safe-haven demand for Bunds, amplifying Spain-Germany spread widening beyond what Spanish fundamentals alone warranted (De Santis [2012](#); De Grauwe and Ji [2013](#)). The extension of SMP purchases to Spanish and Italian government bonds on 7 August 2011 was therefore the most directly relevant ECB intervention for the Spain-Germany spread analysis, marking the first occasion on which Spanish spreads were explicitly targeted by ECB policy. The programme was effective in the short-term, with Spanish bond yields falling almost 100 basis points on the first day alone Ghysels et al. ([2017](#)), though as with the earlier SMP operations, effects remained partial in the absence of an unconditional fiscal backstop.

8 December 2011: Three-year LTROs. Long-Term Refinancing Operations (LTROs) are ECB liquidity facilities through which commercial banks can borrow from the central bank at a fixed rate against eligible collateral for a defined term. The ECB's announcement of an exceptionally long three-year facility at a time when wholesale bank funding markets had become severely impaired, followed by two allotments on 21 December 2011 and 29 February 2012, injected approximately €1 trillion into the Eurozone banking system (European Central Bank [2012a](#)). Following these operations, peripheral banks (including Spanish institutions) substantially increased their domestic sovereign bond holdings, using cheap central bank liquidity to fund a carry trade between ECB borrowing rates and higher peripheral yields (Acharya and Steffen [2015](#)). This made domestic banks the marginal buyers of Spanish government debt and produced a meaningful, if temporary, compression of Spain-Germany spreads in early 2012, though it ultimately deepened the sovereign-bank nexus at the heart of the crisis by adversely affecting bank balance sheets once bond returns deteriorated.

27 June 2012: Spain requests financial assistance for its banking sector. Spain's formal request for EU financial assistance following months of banking sector stress and a surge in ten-year Spanish yields to over 7 percent, is the most significant domestic event for Spanish sovereign spreads in this period. The request illustrates the doom loop mechanism documented by Acharya, Drechsler, and Schnabl ([2014](#)): bank bailouts transfer financial sector losses onto the sovereign balance sheet, raising sovereign credit risk, which in turn erodes the value of banks' government bond holdings, feeding further stress back into the financial sector. As Lane ([2012](#)) notes, this adverse feedback between publicly funded bank recapitalisation and sovereign risk is particularly

acute when the sovereign's own fiscal position is already under pressure: precisely the situation Spain faced in mid-2012.

26 July 2012: Draghi's "whatever it takes" speech. Draghi's speech at the Global Investment Conference in London in which he stated the ECB was ready to do "whatever it takes to preserve the euro", is universally recognised as the turning point of the crisis (Wanke 2017). Within weeks, Spanish bond yields declined substantially (Alcaraz et al. 2019), as markets interpreted this as the ECB becoming the de facto lender of last resort for euro-sovereigns as they could theoretically create unlimited liquidity (Whelan 2022). By making this credible commitment, Draghi eliminated the self-fulfilling dynamics that had pushed spreads beyond warranted levels. The speech has also been directly linked to a reduction in redenomination risk: the fear that peripheral sovereign bonds could be redenominated into devalued domestic currencies (Krishnamurthy, Nagel, and Vissing-Jorgensen 2018). European Parliament (2022) confirms that the speech alone, without the OMT ever being activated, was sufficient to remove redenomination risk from sovereign bond markets.

6 September 2012: OMT technical features announced. The announcement of the OMT programme's technical features translated Draghi's verbal commitment into a concrete policy framework as it permitted potentially unlimited secondary market purchases of sovereign bonds (European Central Bank 2012b). Despite never being activated, the OMT eliminated the structural conditions for panic-driven spread widening: with an unlimited backstop in place, the bad equilibrium in which investor panic forces a solvent sovereign to default was no longer a credible scenario. Following the announcement there was a significant reduction in Spanish spreads (Altavilla, Giannone, and Lenza 2014) via its effects on both default and redenomination risk channels (Krishnamurthy, Nagel, and Vissing-Jorgensen 2018).

The final period and end of the crisis. There is no single official date marking the end of the euro-area sovereign debt crisis. Following the ECB chronology, we treat the acute crisis phase as running from May 2010 to the second half of 2013, with 31 July 2013 used as the empirical cutoff date (Hobelsberger, Kok, and Mongelli 2022). By July 2013, Spanish spreads had declined substantially from their July 2012 peak and no longer exhibited the panic-driven dynamics of the acute crisis phase, making this a natural empirical endpoint for the analysis. However, this does not imply that all country's programmes had ended by mid-2013, for instance, Spain's financial-sector assistance programme continued up until January 2014 (International Monetary Fund 2014).

4.3 Date Categorisation

Though we have identified the dates above as critical events in the Eurozone sovereign debt crisis, for the subsequent stage of our analysis we proceed to categorise them into five discrete episodes, as set out in the table below.

Several of the dates identified above occurred in close proximity and were driven by a common underlying shock. For instance, the two Greek fiscal events form part of a single bond market shock, while the events of summer 2012 spanning Spain's banking sector request, Draghi's speech, and the OMT announcement collectively represent one fragmentation episode. Treating each date as an independent event would risk

Table 1: Consolidation of Euro-area Sovereign Stress Episodes and Anchor Dates

Episode	Dates included	Main anchor date	Reason
Greek fiscal shock	16/10/2009; 05/11/2009	16/10/2009	Same disclosure and repricing episode.
First euro-area intervention	10/05/2010	10/05/2010	Standalone launch of the Securities Markets Programme.
Spain/Italy contagion	07/08/2011	07/08/2011	Standalone expansion of the Securities Markets Programme to Spain and Italy.
LTRO liquidity episode	08/12/2011; 21/12/2011; 29/02/2012	08/12/2011	Same three-year LTRO policy package.
Spanish banking sector assistance	27/06/2012; 26/07/2012; 06/09/2012	27/06/2012	Same summer 2012 fragmentation episode.

Notes: LTRO refers to long-term refinancing operations; OMT refers to Outright Monetary Transactions.

attributing distinct regime shifts to what are simply manifestations of the same structural disruption, introducing artificial clustering into our framework. By consolidating related dates around a single anchor event within each episode, we reduce this noise while preserving the economic content of each transition. The anchor date within each episode is selected on the basis of which event most plausibly constitutes the primary structural break (typically the first public signal) with the remaining dates treated as confirmatory within that same window.

Having established the five discrete episodes and their anchor dates, it is natural to ask how well this hand-selected chronology aligns with what the data itself suggests about structural breaks in spread dynamics. One way to assess this is to compare the selected dates against the regime transitions implied by the Markov-switching model estimated in Section 3. If the historically motivated dates correspond to periods in which the statistical behaviour of the spread genuinely changed, the model should identify transitions in their vicinity. The following section makes this comparison explicit.

4.4 Statistical Date Comparison

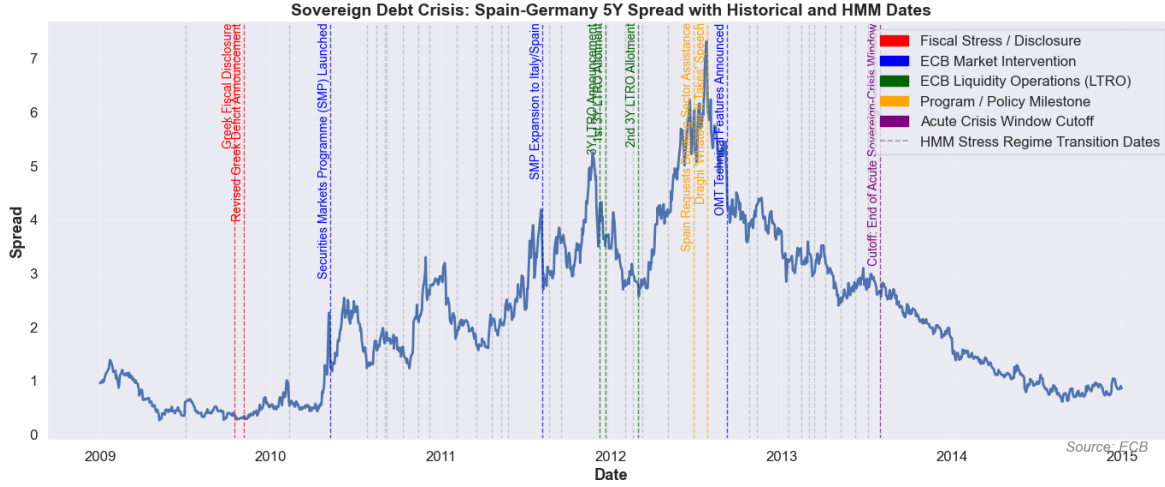


Figure 4: Sovereign Debt Crisis with Historical and HMM Dates.

Table 2: Comparing Historical and HMM Dates

Event date	Historical event name	HMM transition	HMM gap days
2009-10-16	Greek Fiscal Disclosure	2009-07-03	-105
2009-11-05	Revised Greek Deficit Announcement	2010-02-11	98
2010-05-10	Securities Markets Programme (SMP) Launched	2010-07-27	78
2011-08-07	SMP Expansion to Italy/Spain	2011-08-22	15
2011-12-08	3Y LTRO Announcement	2012-02-01	55
2011-12-21	1st 3Y LTRO Allotment	2012-02-01	42
2012-02-29	2nd 3Y LTRO Allotment	2012-03-08	8
2012-06-27	Spain Requests Banking-Sector Assistance	2012-05-03	-55
2012-07-26	Draghi 'Whatever It Takes' Speech	2012-05-03	-84
2012-09-06	OMT Technical Features Announced	2012-10-24	48
2013-07-31	Cutoff: End of Acute Sovereign-Crisis Window	2013-07-05	-26

Notes: HMM gap days denotes the number of calendar days between the historical event date and the nearest HMM stress-regime transition date; negative values indicate that the HMM transition preceded the historical event, while positive values indicate that it followed it.

Table 2 compares the historically selected sovereign-crisis dates with the stress-regime transition dates implied by the Markov-switching model. The coloured vertical lines in Figure 4 indicate the economically selected event dates, while the grey dashed lines show HMM-implied transitions into the stress regime. The purpose of this comparison is not to impose a one-to-one correspondence between historical announcements and statistical regime changes, but to assess whether the data-driven regime classification broadly aligns with the our crisis chronology.

Overall, the HMM stress-transition dates provide partial statistical support for the hand-selected chronology. Several stress-regime transitions occur close to the major sovereign-crisis episodes, especially around the Spain/Italy contagion episode in August 2011, the LTRO period in early 2012 and the OMT-related period in autumn 2012 (see

also Figure 17 in Appendix A.5). For example, the HMM identifies a stress transition shortly after the SMP expansion to Italy and Spain, close to the second LTRO allotment, and near the later cooling phase. This suggests that the selected historical dates correspond not only to economically meaningful events, but also to periods in which the statistical behaviour of the Spain-Germany spread changed.

At the same time, the correspondence is not exact. The Greek fiscal-disclosure dates and the initial SMP announcement are not accurately matched by HMM stress transitions. This is not surprising, since the Markov-switching model identifies changes in the statistical dynamics of the spread, whereas the selected dates correspond to specific fiscal announcements or policy interventions. Some policy dates may therefore occur after stress has already begun, while other HMM transitions may capture earlier market repricing before an official announcement. Following the caution in van den End (2019), the HMM dates are therefore interpreted as a complementary statistical cross-check rather than as definitive crisis dates.

Taken together, the visual comparison and nearest-transition table support the use of the hand-selected dates at the broader episode level. The HMM does not replace the historical chronology, but it strengthens the interpretation that the selected sovereign-crisis episodes are associated with observable changes in the Spain-Germany 5-year spread process.

5 Results and Discussion

The results for each of the five episodes are presented below. The analysis focuses on whether the composite EWS score exhibited meaningful activity in the days and weeks preceding each anchor date, and whether any signals were particularly concentrated in the immediate run-up to the event. For each episode, the accompanying charts plot Spain-Germany sovereign spreads alongside the number of EWS observed leading up to the crisis date. The spread is marked with a red dot for dates with an EWS score of three or over, while the crisis date itself is indicated by a red vertical line. Each figure covers a window from one year before to six months after the event. To avoid mechanical bias, the crisis date is excluded from the EWS calculations.

Presence of EWS Warning Before Crisis Episodes						
Crisis episode	Greek fiscal disclosure	Yes	No	No	No	No
	First ECB sovereign-market intervention - SMP launch	Yes	Yes	Yes	Yes	Yes
	Spain/Italy contagion and SMP expansion	Yes	Yes	Yes	Yes	Yes
	3Y LTRO liquidity episode	Yes	Yes	Yes	Yes	No
	Spanish banking-sector assistance	Yes	Yes	Yes	Yes	Yes
		100d	50d	25d	10d	5d
Pre-event window						

Figure 5: Presence of EWS Warning Before Crisis Episodes.

Before examining each episode individually, Figures 5 and 6 provide a structured overview of EWS behaviour across all five episodes and multiple pre-event horizons, which helps orient the episode-level discussion that follows. Each cell in Figure 5 indicates whether at least one day with three or more simultaneously active EWS indicators was recorded within the given pre-event window. The threshold of three or more concurrent indicators is used throughout as the criterion for a meaningful composite warning, following the logic that co-movement across multiple indicators provides stronger evidence of systemic fragility than any single measure in isolation.

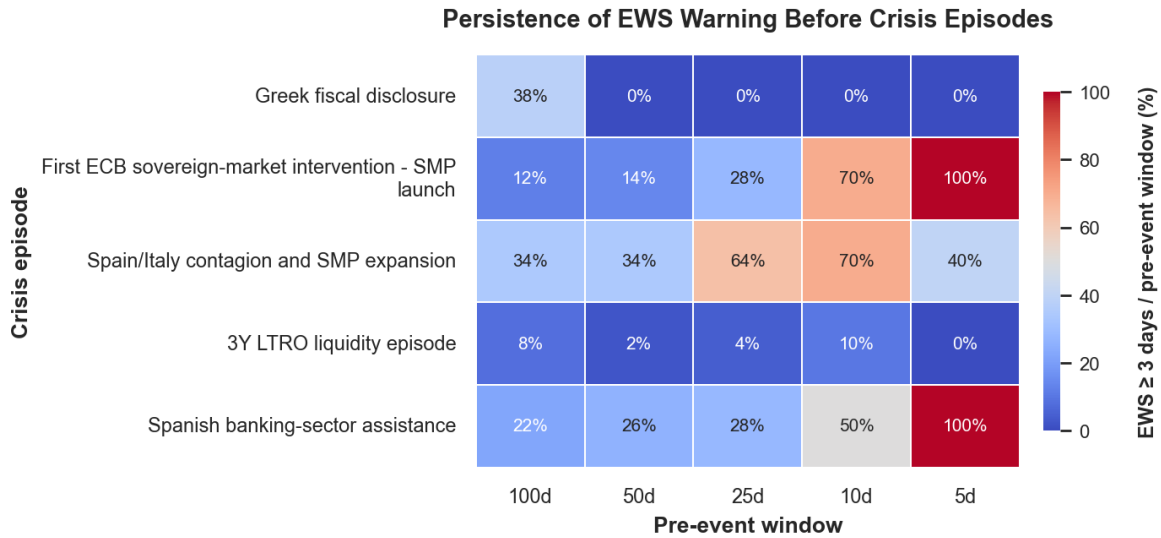


Figure 6: Persistence of EWS Warning Before Crisis Episodes.

Expanding on this, each cell in Figure 6 reports the percentage of trading days within the given pre-event window on which three or more EWS indicators were simultaneously active. Redder shading indicates a higher proportion of flagged days, reflecting more sustained rather than merely episodic signalling. This persistence dimension complements Figure 5 by providing a more detailed view on EWS evolution across each time window.

Whereas Figure 5 requires only a single qualifying day within a window to record a "Yes", meaning a positive result may reflect only very sparse signalling, Figure 6 captures whether EWS activity was genuinely building ahead of an episode. The two figures should therefore be interpreted together throughout the results that follow.

5.1 Greek Fiscal Shock

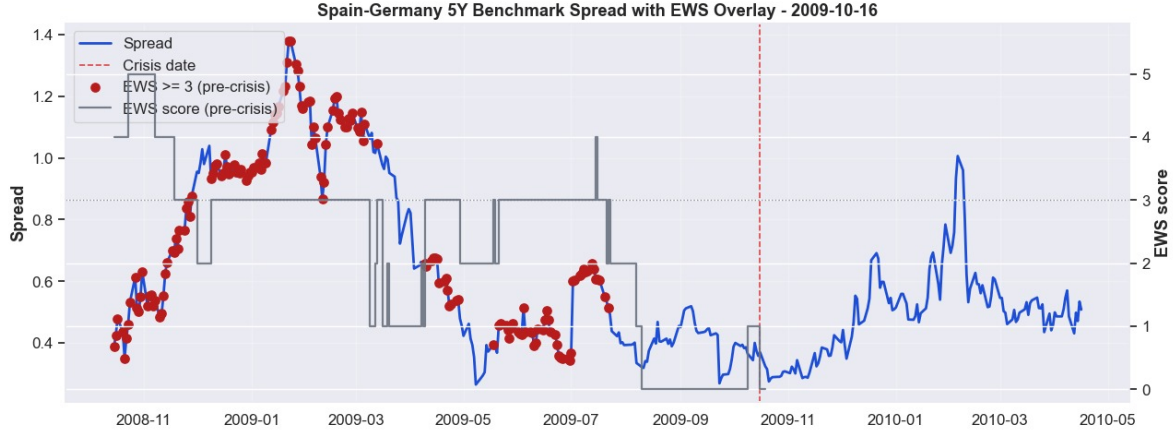


Figure 7: Greek Fiscal Shock.

In the year leading up to the Greek fiscal shock there was substantial variability in spreads. They peaked at nearly 1.4 around eight months prior to the crisis but subsequently narrowed to around 0.4 as they approached the anchor date of 16th October 2009, suggesting that markets had in fact grown calmer and were pricing in relatively little sovereign risk in the immediate lead-up to the disclosure.

The EWS indicators produced very limited signalling activity in the period preceding this event. Although there was a composite EWS score of over three on 38 days in the early part of the 100-day window, the system went entirely quiet in the 50-days immediately before the anchor date, with scores of zero recorded across most of that period and a score of only one on the day of the fiscal disclosure itself.

Rather than representing a failure of the methodology, this is an informative result: the near-total absence of EWS activity suggests that the bond market was not yet approaching a state of instability at this point. This is consistent with the observed narrowing of spreads in the run-up to the event; the market had not priced in a regime shift, and the EWS were not signalling one. This can be easily explained: the Greek fiscal shock was not, in the first instance, a Spanish spread event; it was a disclosure that triggered a broader reassessment of euro-area sovereign risk over the following months. The absence of EWS activity here is therefore perhaps best understood not as a miss but as a meaningful baseline: Spanish spreads were not yet in a pre-transitional state, and the indicators reflected this accurately.

5.2 First Eurozone Intervention

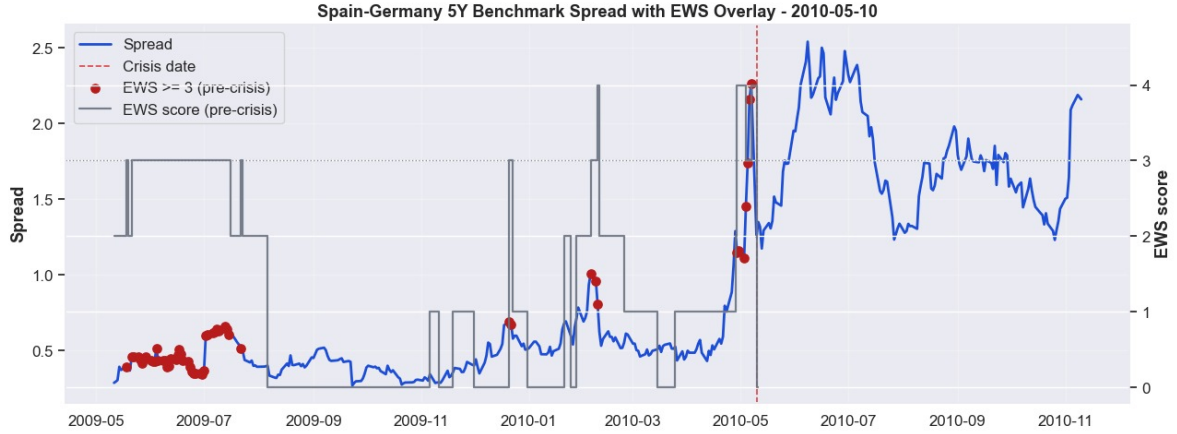


Figure 8: First Eurozone Intervention.

In the months preceding the ECB's first Securities Markets Programme intervention, spreads peaked at around 1.0, captured by a brief but sharp spike in the EWS score for a handful of days. Following this peak, spreads flattened and remained relatively stable at around 0.5 for roughly a quarter. Then, in the month immediately before the anchor date, spreads surged sharply, reaching approximately 2.25 on the day of the intervention and remaining substantially elevated for the following six months.

The EWS indicators produced considerably more meaningful activity ahead of this event than the previous. For seven of the ten days preceding the 10th of May 2010, the composite EWS score exceeded three, including all five of the days immediately before the anchor date. On the majority of these dates, the score exceeded four; a sharp jump from a score of just one only days earlier. The concentration and sudden intensification of signals in the final days of the window is consistent with what complexity theory would predict when a system is approaching a critical transition: statistical indicators such as autocorrelation and variance tend to spike as the system loses resilience and nears a tipping point. The EWS were therefore signalling that the spread system was in a state of instability in the days before the intervention and that a regime shift was imminent. It is worth noting that over the broader 100-day window, EWS activity was sparse: scores of three or more appear on only twelve days in total, with seven of those concentrated in the final ten days. This suggests that the instability was rapid in onset rather than a long-building process.

5.3 Spain Contagion

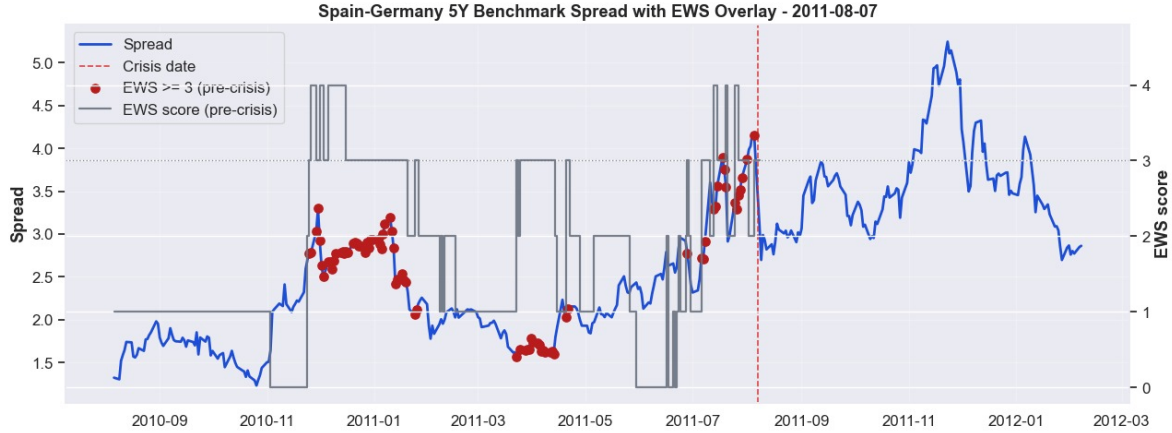


Figure 9: Spain Contagion.

Throughout this one-and-a-half-year period, spreads followed a general upward trend, rising from around 1.5 in September 2010 to 3.0 by March 2011. Towards the end of 2010, spreads experienced a sudden increase, almost doubling to 3.0 in a matter of weeks. This was accompanied by a marked rise in EWS activity, with the composite score exceeding three for a sustained period and at points reaching four: signalling that the system was in a highly unstable state during this episode. Following this, spreads became somewhat more stable and settled within a corridor of roughly 1.5-3.0 through to July 2011, though EWS flags continued to appear intermittently, mostly concentrated around April.

From July onwards, spreads began rising again, breaking above the corridor and reaching approximately 4.0 in the days immediately preceding the anchor date. This was accompanied by increasing variability in the EWS composite score, which oscillated up and down while following a general upward trajectory. This behaviour is itself one of the theoretical indicators of an approaching critical transition, and its presence here is consistent with the methodology.

In the ten days immediately preceding the event, the composite EWS score exceeded three on seven out of ten days, with two of these occurring in the final five days. This is a somewhat weaker concentration than was observed ahead of the SMP launch, but the broader picture across longer horizons is considerably stronger. Over the 25- and 50-day windows, the EWS score exceeded three on 16 and 17 days respectively, and over the full 100-day window the figure stands at 34 days. The EWS were therefore signalling the presence of an unstable system and an approaching regime change across multiple time horizons ahead of the SMP expansion to Spain. This is consistent with the fact that the Spanish contagion was a more endogenous development in Spanish sovereign spreads than the Greek fiscal shock, making it precisely the kind of event this methodology is best placed to detect.

5.4 LTRO

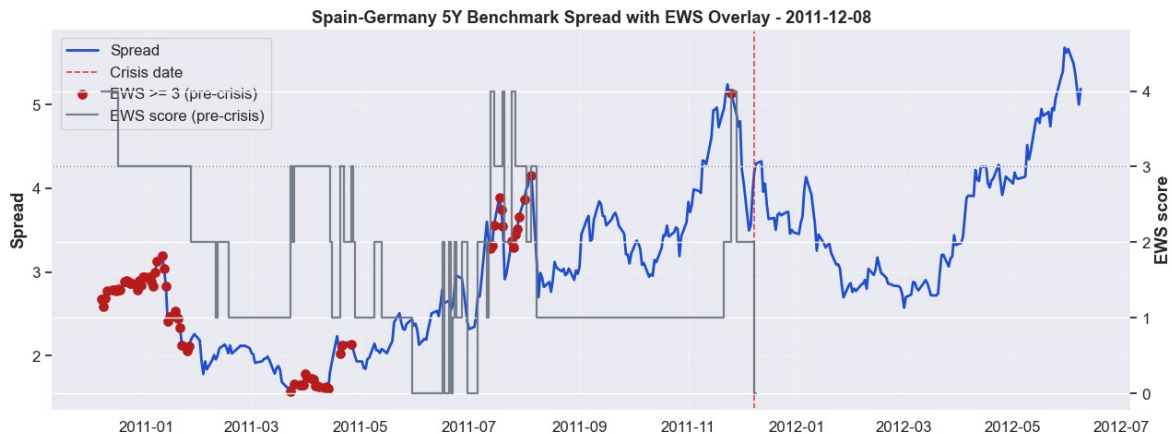


Figure 10: LTRO.

Following the Spanish contagion and the expansion of the SMP, spreads fell sharply from their peak of around 4.0 to below 3.0, coinciding with a correspondingly rapid decline in the EWS composite score to one, where it remained largely flat for approximately three months. A modest uptick in spreads around September was quickly reversed.

From November, however, spreads began rising again, briefly exceeding 5.0 for the first time in this period. This was accompanied by a short-lived single-day spike in the EWS score to four. Between this date and the LTRO announcement, spreads fell back to around 3.5 and EWS activity remained very subdued, reaching zero on the final observation before the event.

Across all time windows, the EWS system produced very few signals ahead of the LTRO. Over the full 100-day window, only eight days registered a score above three, with just one of these occurring in the final ten days. This near-total absence of EWS activity suggests that Spanish spreads were not in a state of critical instability in the build-up to the LTRO announcement. Rather, the market appears to have been in a relatively contained, if elevated, state, with the LTRO simply representing an institutional policy intervention independent of any critical transition. The EWS methodology does not claim to anticipate all market-moving events, only those accompanied by the measurable loss of system resilience that precedes a regime shift, which was absent on this occasion.

5.5 Spanish banking-sector assistance

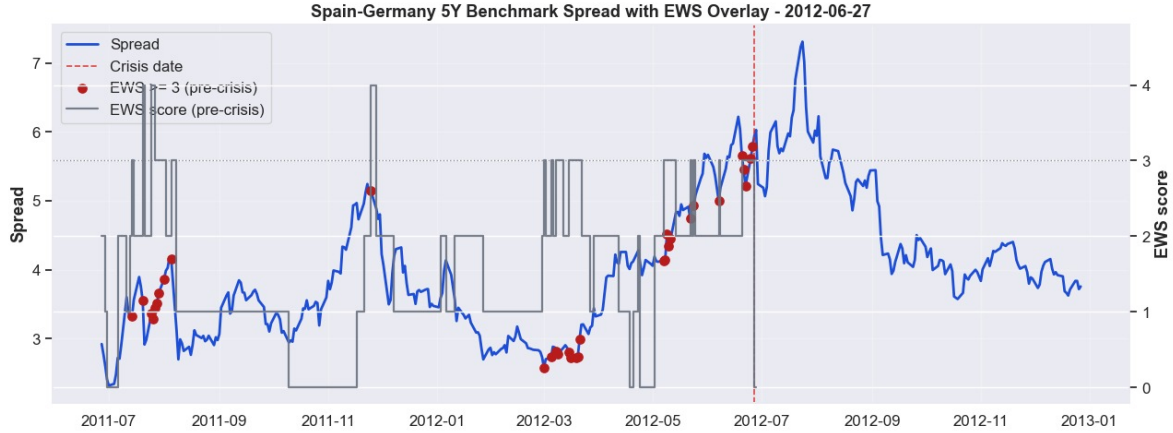


Figure 11: Spanish banking-sector assistance.

Spreads gradually declined in the months following the LTRO introduction, reaching a low of around 2.5 in March 2012, accompanied by relative stability in the EWS indicators. Notably, it is at precisely this low point that the EWS begin to register renewed signals of instability, with a cluster of dates showing scores exceeding three. This was followed by a steady increase in spreads over the subsequent months, lending support to the indicators' capacity to flag emerging instability ahead of a deterioration.

As the anchor date approached, the signalling became more consistent: on all five of the days immediately preceding the event, the EWS composite score stood at three. This concentrated activity in the final days is precisely what complexity theory would associate with a system on the brink of a critical transition; spread dynamics had become sufficiently unstable that a regime shift was increasingly likely. The EWS were therefore indicating, in the days before Spain's formal request, that the system had entered a highly unstable state. They did not predict the political decision itself, but did accurately reflect the underlying fragility of the spread regime.

Following Spain's request for financial assistance, spreads experienced a brief sharp fall before widening again to a peak of over 7.0 around late July, prior to Draghi's "whatever it takes" speech. That intervention reassured markets of the ECB's commitment, and spreads subsequently fell, settling at around 4.0 roughly six months after Spain's request.

Taken individually, each of the five episodes reveal a distinct relationship between EWS activity and the nature of the underlying shock; the patterns that emerge from this comparison are discussed in the conclusion that follows. Before turning to that broader assessment, however, we first examine whether EWS dates carry statistically detectable information about subsequent spread dynamics, using a formal hypothesis test to assess whether the indicators identify periods that are systematically different from randomly selected dates in terms of future spread behaviour.

5.6 Spread Dynamics Following EWS Dates

Table 3: Welch’s T-test Results: EWS Signals vs Random Future Spread Changes

Horizon	N signal	N random	Signal mean	Random mean	Signal sd	Random sd	Mean diff	t	p-value
spread_change_5d_after	130	130	-0.01	0.02	0.36	0.11	-0.03	-0.78	0.4388
spread_change_10d_after	130	130	-0.08	0.03	0.43	0.16	-0.11	-2.68	0.0080
spread_change_15d_after	130	130	-0.08	0.04	0.60	0.19	-0.12	-2.20	0.0293
spread_change_20d_after	130	130	-0.07	0.04	0.69	0.24	-0.11	-1.79	0.0760
spread_change_25d_after	130	130	-0.10	0.03	0.72	0.24	-0.13	-1.94	0.0542

Before presenting results, we note that some signal dates are inherently clustered and future spread-change windows can overlap, meaning these tests should be interpreted as exploratory rather than as fully independent hypothesis tests. For each forecast horizon $h \in \{5, 10, 15, 20, 25\}$, Table 3 reports Welch’s two-sample t-tests comparing future spread changes after EWS signal dates with future spread changes after randomly selected dates. The null hypothesis is that the mean future spread change is equal across the two groups, while the two-sided alternative allows for a difference in either direction.

$$H_0 : \mu_{\text{signal},h} = \mu_{\text{random},h} \quad (10)$$

$$H_1 : \mu_{\text{signal},h} \neq \mu_{\text{random},h} \quad (11)$$

Where $\mu_{\text{signal},h}$ denotes the average h -day future spread change following EWS signal dates, and $\mu_{\text{random},h}$ denotes the average h -day future spread change following randomly selected dates.

The results do not provide evidence that EWS signal dates are followed by stronger subsequent spread widening. At the 5-day horizon, the difference between signal and random dates is small and statistically insignificant. At the 10- and 15-day horizons, the null of equal mean changes is rejected at the 5% level, but the estimated mean differences are negative. This means that future spread changes after EWS signal dates are, on average, lower than after random dates. The 20- and 25-day horizons are also negative and marginally significant at approximately the 10% level.

Economically, this suggests that the EWS score should not be interpreted as a short-horizon directional predictor of further spread widening. Instead, the signal dates appear to occur during periods in which spreads are already elevated or unstable, after which spreads often stabilise or partially narrow. This interpretation is reinforced by the larger standard deviations in the signal group, which indicate that spread changes following EWS signals are more dispersed than those following random dates. Overall, the test supports the interpretation of the EWS score as a marker of elevated sovereign-spread stress or fragility, rather than as a direct predictor of additional spread widening over the following 5 to 25 trading days.

However, this result is also consistent with the broader interpretation of complexity-based early-warning indicators. The purpose of the EWS score is not necessarily to predict that spreads will continue widening mechanically after a signal date. Rather, the indicators are designed to capture moments in which the system becomes unstable, fragile, or less resilient. Once such instability becomes visible, spreads may subsequently narrow not because the underlying stress disappears endogenously, but because policy

intervention occurs. In the euro-area context, this may include actions or announcements by the ECB that stabilise sovereign bond markets and compress spreads. Therefore, the finding that spreads often stabilise or narrow after EWS signal dates does not necessarily weaken the complexity-theory interpretation. Instead, it suggests that the indicators can successfully identify periods of elevated instability, after which external intervention or market stabilisation mechanisms may lead to spread compression.

6 Conclusion

This thesis set out to ask whether complexity-theory early-warning signals can provide meaningful advance notice of regime shifts in euro-area sovereign bond markets. The answer is cautiously affirmative. Across four of the five episodes examined, the EWS composite score registered meaningful activity ahead of the anchor date. The Greek fiscal shock produced near-total silence from the indicators, interpretable not as a methodological failure but as an informative baseline, since Spanish spreads were not yet in a pre-transitional state at that moment. The LTRO episode similarly produced very limited EWS activity, though for a different reason: the intervention was a discretionary policy decision rather than the culmination of endogenous spread fragility, and the indicators correctly reflected the absence of a critical transition in the build-up to it. The distinction between endogenous fragility and exogenous shock is central to determining what the EWS framework can and cannot be expected to detect.

The statistical analysis of spread dynamics following EWS dates reinforces the core finding: the indicators function as markers of elevated systemic fragility rather than directional predictors, identifying when a system is unstable and susceptible to a transition rather than forecasting whether the next move will be an escalation or a policy-driven compression. The Markov-switching model provides partial but imperfect corroboration of the hand-selected chronology, and its limitations (near-identical AR(1) coefficients across regimes and residual ARCH effects) point toward richer model structures as a natural next step. A further extension would be to disaggregate the composite EWS score and examine which specific indicators are driving signals ahead of each episode. This would clarify whether certain indicators consistently lead others, and whether different types of shock activate different subsets of the framework.

The broader contribution of this thesis lies less in its specific empirical findings than in what those findings suggest about the transferability of the complexity-theory framework. van den End (2019), working with euro-area interest rates, found clean and consistent early-warning signals in a relatively liquid, institutionally stable market. The Spain-Germany sovereign spread is a considerably more challenging test case: it is noisier, subject to abrupt policy interventions, and shaped by forces that have no direct analogue in the interest rate context (fiscal shocks, redenomination risk, contagion). That the EWS framework produces interpretable and theoretically coherent results even here suggests its applicability may be wider than the existing literature implies. If these indicators can detect fragility in something as institutionally complex and politically contingent as peripheral sovereign spreads during the euro-area debt crisis, there is a reasonable case that they can do so elsewhere.

The most immediate extension is the application of this framework to other sovereign

spread pairs and, beyond fixed income, to other asset classes or currency pairs where regime dynamics are economically meaningful but difficult to anticipate. Within the euro area, Portugal and Italy offer natural comparators: each experienced a distinct crisis trajectory, and variation in EWS performance across this cross-section would clarify whether the indicators are capturing a common systemic dynamic or something more idiosyncratic to Spanish spread behaviour. Outside the euro area, the framework could be applied to emerging market sovereign spreads, where tipping-point dynamics and self-fulfilling crises are if anything more pronounced, or to currency pairs in managed exchange rate regimes, where the build-up to a devaluation or peg abandonment has precisely the character of a critical transition the methodology is designed to detect.

A further direction concerns the relationship between EWS signals and institutional responses. One of the more striking patterns in the results is that spread stabilisation or compression frequently follows EWS activity. This is not because the indicators are wrong, but because elevated fragility tends to prompt intervention. This raises a question that the current framework cannot answer: to what extent do policymakers, implicitly or explicitly, respond to the same market signals that trigger the EWS? And conversely, as these tools become more widely known, could systematic monitoring by sophisticated institutional investors cause fragility to be precipitated before a mathematical tipping point is reached? These questions sit at the intersection of complexity theory, institutional economics, and market microstructure, and they represent the most intellectually interesting frontier this work opens up.

A Markov-Switching Autoregressive Model

A.1 Suitability of the Spain-Germany 5-Year Spread for Regime-Switching Analysis

Before estimating the regime-switching model, the distributional properties of the Spain–Germany 5-year benchmark yield spread are examined. The spread is defined as:

$$s_t = y_{t,5Y}^{ES} - y_{t,5Y}^{DE}, \quad (12)$$

where $y_{t,5Y}^{ES}$ is the Spanish 5-year benchmark yield and $y_{t,5Y}^{DE}$ is the German 5-year benchmark yield. A positive value of the spread means that Spanish 5-year sovereign debt trades at a higher yield than German 5-year sovereign debt. In this context, a higher spread is interpreted as a sign of higher perceived Spanish sovereign risk relative to Germany.

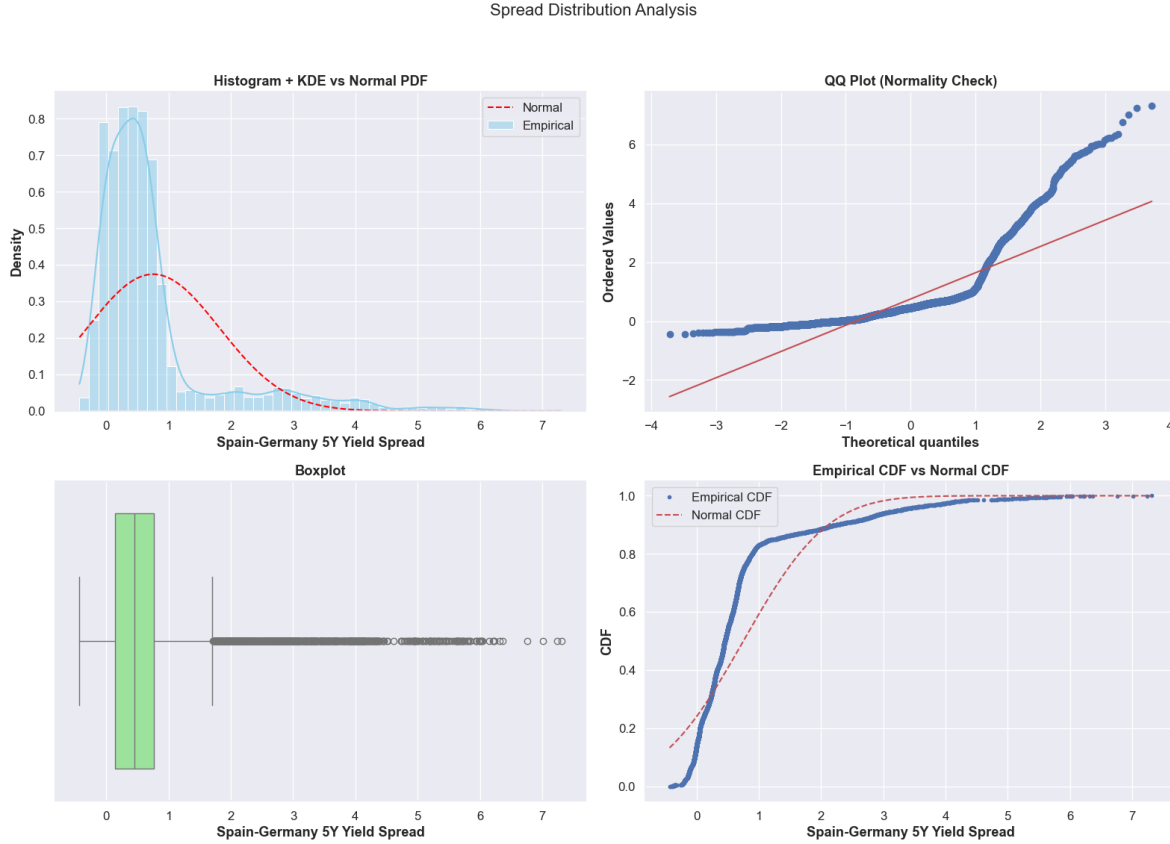


Figure 12: Distributional analysis of the Spain-Germany 5-year benchmark yield spread.

The first diagnostic figure compares the empirical distribution of the spread with a normal distribution. The histogram and kernel density estimate show that most observations are concentrated at relatively low spread levels, while a smaller number of observations are located far in the right tail. This indicates that the distribution is asymmetric. The right tail reflects episodes in which the Spanish yield rose strongly relative to the German yield, which can be interpreted as periods of increased sovereign stress.

The QQ plot confirms that the spread is not normally distributed. If the spread followed a normal distribution, the points in the QQ plot would lie close to the straight reference line. Instead, the upper tail bends clearly away from the line, showing that large spread values occur more frequently than would be expected under normality. In the sample, the spread has a skewness of 2.4369 and excess kurtosis of 6.3778, which indicates a positively skewed and fat-tailed distribution.

The boxplot gives a similar conclusion. Most observations lie within a relatively narrow range, but there are many positive outliers. In this application, these outliers should not be interpreted as data errors. They are economically meaningful because they correspond to periods in which investors demanded a larger yield premium for holding Spanish sovereign bonds relative to German Bunds.

The empirical cumulative distribution function also differs from the normal cumulative distribution function. The empirical distribution rises quickly at low spread levels, showing that a large part of the sample is concentrated in calmer market conditions. However, the right tail is much heavier than under a normal distribution. Overall, these diagnostics suggest that the Spain-Germany 5-year spread is not well described by a single normal distribution. Instead, the data are more consistent with a mixture of different states, such as a low-spread calm regime and a high-spread stress regime. This provides us with an initial motivation for estimating a two-state regime-switching model.

A.2 Full-Sample ADF Test

To examine the persistence of the spread, we applied an Augmented Dickey-Fuller (ADF) test to the level of the Spain-Germany 5-year spread. The null hypothesis of the ADF test is that the series has a unit root and is therefore non-stationary. Therefore, the alternative hypothesis is that the series is stationary. A stationary time series has a stable long-run structure. Its mean, variance, and autocorrelation pattern do not systematically change over time. The series may move up and down, but it fluctuates around a relatively stable level. Shocks are usually temporary, so after a disturbance the series tends to return toward its normal range. On the other hand, a non-stationary time series does not have a stable long-run structure. Its mean or variance may change over time, or shocks may have permanent effects. The series may drift upward or downward, follow a trend, or move through different regimes. In this case, past shocks can permanently change the level of the series.

The full-sample ADF statistic is equal to -2.2832 , with a p-value of 0.1774. The critical values are -3.4313 at the 1% level, -2.8620 at the 5% level, and -2.5670 at the 10% level. Since the test statistic is not more negative than any of these critical values, and since the p-value is above conventional significance levels, the unit-root null hypothesis cannot be rejected.

This result suggests that the spread is highly persistent in levels. In economic terms, shocks to the Spain-Germany spread do not disappear immediately. When the spread widens during periods of market stress, it can remain elevated for an extended period. This finding supports the use of an autoregressive regime-switching specification rather than a simple Gaussian hidden Markov model, because the model should account for the dependence of today's spread on its previous values.

At the same time, this result should be interpreted carefully. Sovereign spreads can contain structural breaks and regime shifts, especially around crisis periods. A

Table 4: ADF Test on Spain-Germany 5Y Yield Spread

Metric	Value
ADF Statistic	-2.2832
p-value	0.1774
Critical Value (1%)	-3.4313
Critical Value (5%)	-2.8620
Critical Value (10%)	-2.5670

full-sample ADF test may therefore fail to reject the unit-root null partly because the mean of the process changes over time. For this reason, we use the ADF test as a diagnostic for persistence, not as a final decision rule for model selection.

A.3 ACF and PACF Diagnostics

The autocorrelation function (ACF) and partial autocorrelation function (PACF) are used to examine the time-series dependence of the spread. The ACF measures the correlation between the spread and its own lagged values, while the PACF measures the direct correlation at each lag after controlling for shorter lags.

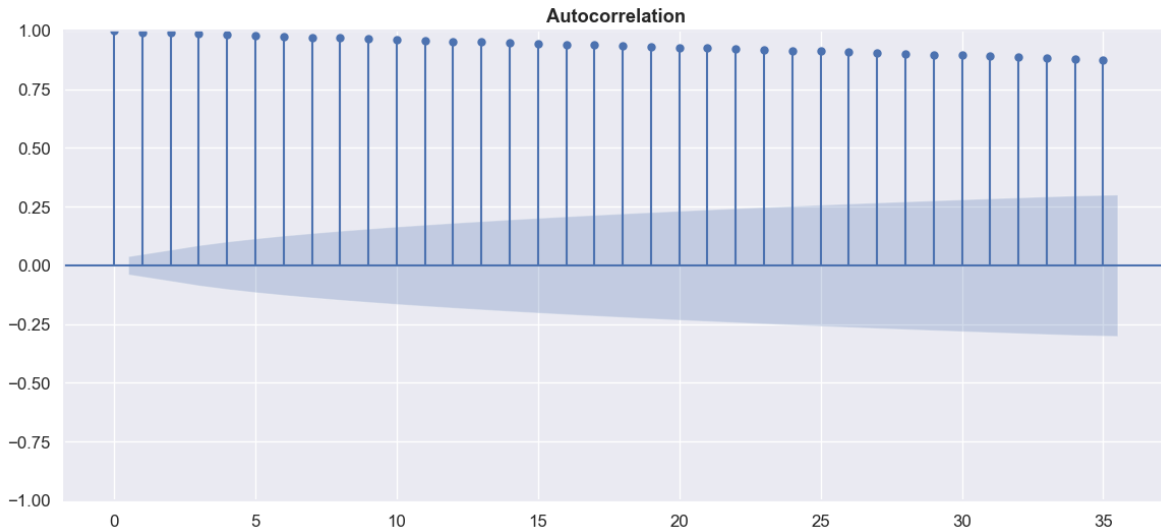


Figure 13: ACF Diagnostic of the Spain-Germany 5-year benchmark yield spread.

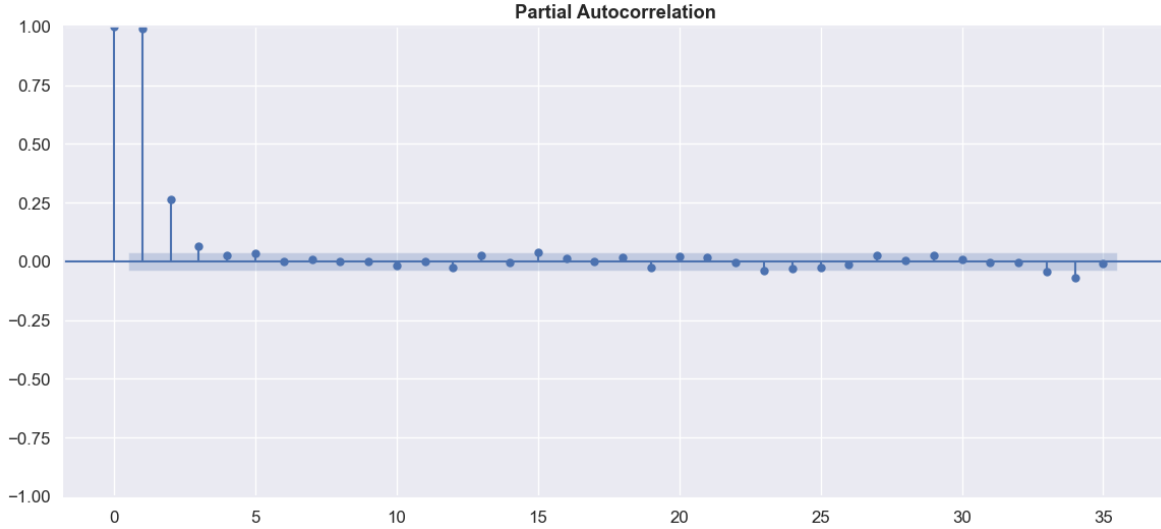


Figure 14: PACF Diagnostic of the Spain-Germany 5-year benchmark yield spread.

The ACF of the spread level shows very high persistence. Autocorrelations remain large over several lags, which means that the spread is strongly related to its past values. This is consistent with the ADF result and indicates that the spread does not quickly return to a fixed mean after a shock. This behaviour can be plausible for sovereign spreads, since market perceptions of credit risk, liquidity risk, and euro-area fragmentation risk tend to adjust gradually rather than instantly.

The PACF shows that the first lag is especially important. This suggests that a low-order autoregressive specification can capture a large part of the short-run dependence in the spread. In particular, the strong first-lag effect supports the use of an AR(1)-type specification as a baseline model.

These results suggest that the spread level contains the relevant long-lived regime information. The ACF and PACF diagnostics also support the choice of a Markov-switching autoregressive model instead of a simple Gaussian HMM. A Gaussian HMM would classify observations by their regime-specific distribution, but it would not explicitly model the strong dependence of the spread on its own lagged values.

A.4 Fitting the HMM AR(1) Model

We have fitted the model between 2009-01-01 and 2013-07-31 on the Spain-Germany 5Y spread. The sample contains 1,195 observations with an overall mean spread of 2.40 and standard deviation of 1.52, ranging from 0.27 to 7.31. The HMM split seems well balanced across regimes (602 vs 593 observations). Regime 1 has a substantially higher mean spread (3.39) than Regime 2 (1.40), consistent with a high-stress vs lower-stress interpretation.

Table 5: Descriptive Statistics of Spain-Germany 5Y Spread and HMM Regimes

Group	Mean	Std. Dev.	Min	Max	Count
Full sample	2.40	1.52	0.27	7.31	1195
Regime 1	3.39	1.23	0.60	7.31	602
Regime 2	1.40	1.06	0.27	4.40	593

A.5 Analysis of Regime Switches

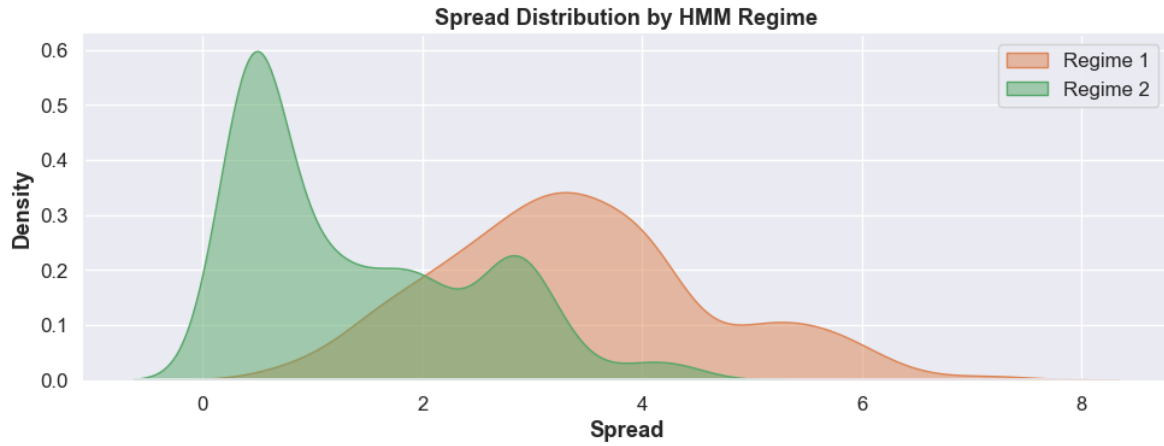


Figure 15: Coloured Spread Distribution, All HMM Regimes.

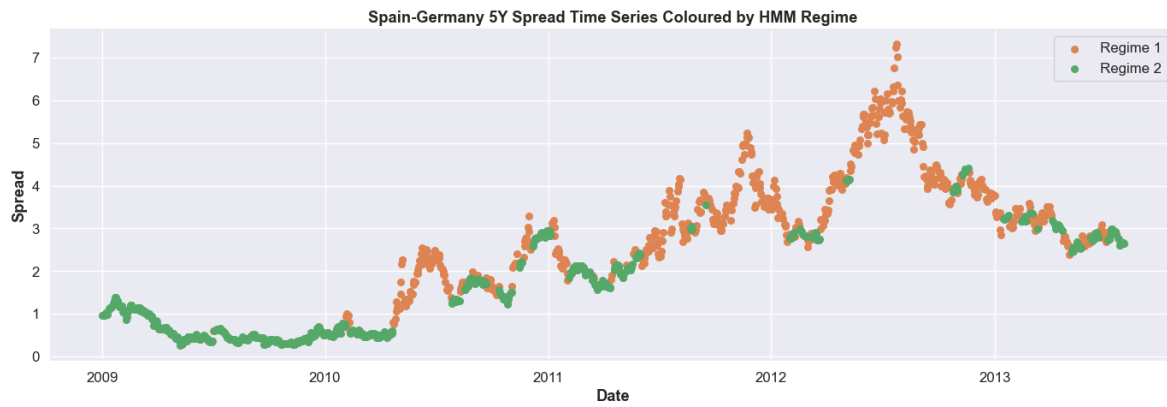


Figure 16: Spain-Germany 5Y Spread Time Series Coloured by HMM Regime.

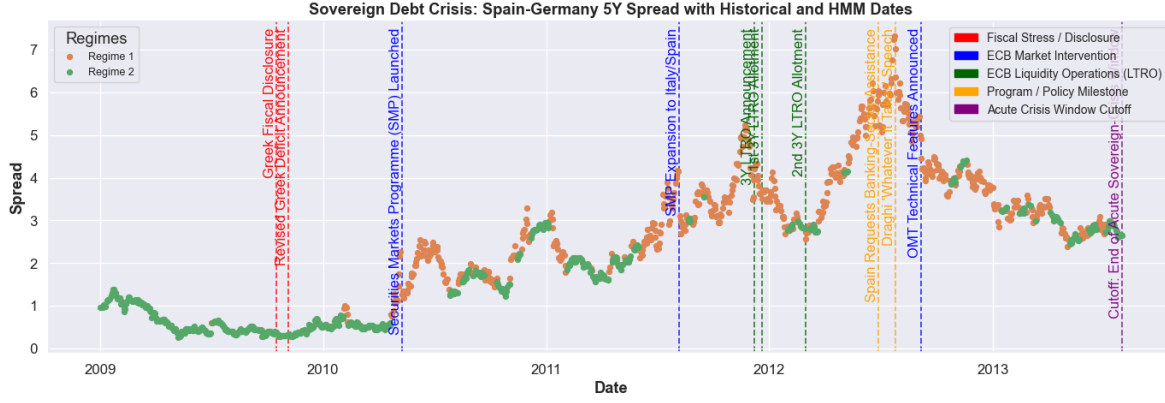


Figure 17: Sovereign Debt Crisis with Historical Dates and HMM Regime Changes.

The regime-coloured spread plots illustrate how the Markov-switching AR(1) model classifies the Spain–Germany 5-year sovereign spread over the crisis sample. In the time-series plot, Regime 1 is concentrated in periods where the spread is elevated and more volatile, especially during the escalation of the euro-area sovereign debt crisis in 2011–2012. By contrast, Regime 2 appears more frequently during lower-spread and comparatively stable periods, particularly before the main crisis escalation and during calmer intervals within the sample.

The distribution plot (Figure 15) reinforces this interpretation. Regime 1 has a distribution shifted clearly to the right, indicating that it is associated with higher spread levels. Regime 2 is concentrated around lower spread values, although there is some overlap between the two regimes.

Overall, the Markov-switching AR(1) model provides a classification of the Spain–Germany spread into a higher-spread stress regime and a lower-spread calm regime. However, the regime classification should not be interpreted as proof of a critical transition by itself. Rather, it is used as a statistical dating tool that complements the hand-selected crisis dates.

Table 6: Estimated HMM Transition Probabilities

	Regime 1	Regime 2
Regime 1	0.9412	0.0569
Regime 2	0.0588	0.9431

To continue, Table 6 reports the estimated transition probability matrix of the two-state AR(1) Markov-switching model. The diagonal elements are high, with $P_{11} = 0.9412$ and $P_{22} = 0.9431$. This means that, conditional on being in a given regime today, the probability of remaining in the same regime on the next day is approximately 94% for both regimes. Conversely, the off-diagonal probabilities are low, around 6%, indicating that transitions between the two regimes occur relatively infrequently.

This suggests that the estimated regimes are persistent rather than short-lived daily fluctuations. The implied expected duration of each regime is approximately $1/(1 - P_{ii})$, which gives around 17 trading days for both Regime 1 and Regime 2. Therefore, the

model contains a meaningful degree of regime persistence, in line with the diagnostic criterion that P_{11} and P_{22} should be close to one. Overall, the transition matrix supports the interpretation that the HMM identifies persistent calm and stress states in the Spain–Germany spread rather than unstable or randomly alternating classifications.

Table 7: Regime Classification and AR(1) Persistence by Regime

Item	Value
Stress regime index	1
Calm regime index	2
AR(1) coefficient in stress regime, ϕ_{stress}	1.000000
AR(1) coefficient in calm regime, ϕ_{calm}	0.999998

Furthermore, the Markov-switching AR(1) model is estimated with regime-specific autoregressive coefficients by setting `switching_ar=True` in the corresponding `Python` function. This avoids imposing equal persistence across calm and stress regimes and makes the regime-switching model consistent with the critical-slowness logic of the EWS analysis, where rising persistence is one of the central mechanisms.

Table 7 reports the the estimated AR(1) coefficients corresponding to each regime. Both estimated coefficients are extremely close to one, with $\phi_{\text{stress}} = 1.000000$ and $\phi_{\text{calm}} = 0.999998$. This indicates that the Spain–Germany 5-year spread is highly persistent in both regimes.

However, the switched-persistence specification does not reveal a meaningful difference between the calm and stress regimes. Although the stress-regime coefficient is marginally higher numerically, the difference is quite insignificant. Therefore, the Markov-switching model does not provide strong direct cross-validation of the critical-slowness claim through higher stress-regime persistence. Instead, it suggests that persistence is high in both states. This tension is important to highlight because the rolling EWS indicators capture local changes in persistence before transitions, while the Markov-switching AR coefficient measures average persistence within each regime.

Table 8: Residual Diagnostics: Ljung–Box and ARCH–LM Tests

Test	Statistic	p-value
Ljung–Box (Q), lag 10	74.20	6.80×10^{-12}
Ljung–Box (Q), lag 20	106.33	9.14×10^{-14}
ARCH–LM	0.0000	< 0.0001

Table 8 reports residual diagnostics for the Markov-switching AR(1) model. The Ljung–Box statistics are large at both lag 10 and lag 20, with p-values of 6.80×10^{-12} and 9.14×10^{-14} , respectively. Since both p-values are far below 5%, the null hypothesis of no residual autocorrelation can be rejected at the 5% significance level. This indicates that the AR(1) structure does not fully absorb the serial dependence in the Spain–Germany 5-year spread. Following this diagnostic result, an AR(2) Markov-switching specification

could be considered, which is also consistent with the earlier PACF evidence suggesting possible second-order dependence.

The ARCH–LM test also rejects strongly, with a p-value below 0.0001. This suggests that residual conditional heteroskedasticity remains even after allowing for regime-specific variances. Therefore, while the Markov-switching model provides an interpretable classification of calm and stress regimes, the residual diagnostics indicate that the baseline AR(1) specification is not fully adequate. The remaining ARCH effects should be acknowledged as a limitation of the model.

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